

Design and Simulation of a Two-Way Traffic Light Control System Using Arduino UNO

Introduction

This mini project presents the design and simulation of a two-way traffic light control system using Arduino UNO in Proteus ISIS Professional. The system controls traffic signals for two directions using predefined timing logic.

Objective

To understand traffic signal automation, learn digital output interfacing, and simulate embedded systems using Proteus.

Software and Tools Used

Arduino UNO, Proteus ISIS Professional, Embedded Systems Concepts

Hardware Components

S.No	Component Name	Specification	Quantity
1	Arduino UNO R3	ATmega328P	1
2	Red LED	5mm	2
3	Yellow LED	5mm	2
4	Green LED	5mm	2
5	Resistors	330 Ω	6
6	Connecting Wires	Jumper Wires	As required

Circuit Description

Six LEDs are used to represent red, yellow, and green signals for two directions. Each LED is connected to the Arduino through current-limiting resistors and properly grounded.

Working Principle

At any given time, one direction is allowed to move while the other is stopped. The traffic signals switch sequentially between red, yellow, and green states with fixed delays, ensuring safe and

orderly traffic flow.

Real-Time Application

Used in road intersections, pedestrian crossings, parking exits, and industrial traffic management systems.

Advantages

Simple design, real-time relevance, easy to understand, suitable for beginners.

Limitations

Fixed timing control and absence of traffic density-based sensing.

Future Enhancements

Density-based control using sensors, pedestrian push buttons, interrupt-based timing, and extension to four-way intersections.

Conclusion

This mini project successfully demonstrates traffic signal automation using embedded systems and provides practical exposure to real-time control logic.